

Statistical Decision Making 45-752

Applying Multiple Regression: CEO Pay

This file is named: 12_slides.pptx
Commands are in: 12_notebook.ipynb

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Industries and Value of Output in the US Economy

Here is an overview of employment and value of production in the US in 2022.

The total number of employed workers was 164.5 million.

The combined value of output was \$45.3 trillion dollars.

- Note: Value of output is different from value added. A retailer sells goods bought from a wholesaler. The value of sales by the retailer is counted as value of output of the retailer. The revenue received by the wholesaler is counted as value of output of the wholesaler. In short, there is double counting of value of output.
- There is not double counting of employment in the table.

Industry Sector	Output 2022 (billions)	Employment 2022 (Thousands)
All industries	45268.90	164482.60
Goods-producing excluding agriculture	9749.80	21133.60
• Mining	776.54	528.95
• Construction	2975.80	7748.06
• Manufacturing	6897.30	12825.71
Service-providing excluding special industries	32987.80	120551.90
• Utilities	724.90	553.09
• Wholesale trade	2596.36	5892.68
• Retail trade	2568.00	15475.48
• Transportation and warehousing	1744.10	6851.19
• Information	2469.86	3074.48
• Financial activities	6397.90	9044.59
• Professional and business services	5431.00	22571.59
• Education services	434.58	3794.78
• Health care and social assistance	2977.00	20555.09
• Leisure and hospitality	1900.00	15835.29
• Other services	839.86	6362.78
• Federal government	1446.40	2909.43
• State and local government	3356.90	19301.89
Agriculture, forestry, fishing, and hunting	583.10	2184.40
Nonagriculture self-employed	Not Available	9112.60

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US Employment in 2024 and 2025

HOUSEHOLD DATA
Summary table A. Household data, seasonally adjusted
[Numbers in thousands]

Category	Dec. 2024	Dec. 2025
Employment status		
Civilian noninstitutional population.....	269,638	274,816
Civilian labor force.....	168,506	171,495
Participation rate.....	62.5	62.4
Employed.....	161,586	163,992
Employment-population ratio.....	59.9	59.7
Unemployed.....	6,920	7,503
Unemployment rate.....	4.1	4.4
Not in labor force.....	101,132	103,321

Source: https://www.dol.gov/newsroom/economicdata/empisit_01092026.pdf

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Firms with Highest Market Cap at End of 2025

Company	Market Cap End-2024	Market Cap End-2025
NVIDIA	2.32	4.60
Apple	2.95	4.00
Alphabet (Google)	2.34	3.80
Microsoft	2.80	3.70
Amazon	1.63	2.60
Saudi Aramco	2.05	1.70
Broadcom	0.60	1.65
Meta Platforms	1.20	1.60
Taiwan Semiconductor (TSMC)	0.95	1.50
Tesla	0.79	1.40
Berkshire Hathaway	0.78	1.10
Walmart	0.43	1.00

Source: Compiled by Microsoft Copilot

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Our Dataset



We now apply statistical analysis to investigate the determinants of CEO Compensation.

I obtained data for all US Firms with more than 99 employees and CEO compensation greater than \$99,000.

Here is a summary overview of these CEO's and their firms in 2022:

- The combined value of the 992 firms in our data, \$24.91 trillion, comprised roughly 62% of total stock market capitalization in the US.
- Together, these firms had 22,648,322 employees, roughly 14% of the US workforce.
- The combined compensation of the CEO's in our data was \$9.437 billion.

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Objectives for This Application



My goals with this application are two-fold.

One objective is to help you become more familiar with data for the US economy.

The other objective is to give you practice in formulating and estimating a model and interpreting the results.

I think you will find this application very interesting.

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Summary Statistics for The Full Sample



The firm with the largest number of employees was Walmart, with 1,541,000 employees.

The CEO with longest service had 43 years as CEO.

Median CEO age was 58.

The 25th percentile of Total Direct Compensation (TDC1) was \$4.545 million.

	TDC1	Employees	MktCap	SIC2	YRS_CEO	AGE
count	989,000	989,000	989,000	989,000	989,000	989,000
mean	9.542	22900.224	25190.335	46.991	8.265	58.217
std	8.329	67358.410	128638.175	18.870	7.341	6.758
min	0.246	105,000	4,582	10,000	1,000	35,000
25%	4.545	2000,000	1772,565	35,000	3,000	54,000
50%	7.498	6298,000	4634,258	48,000	6,000	58,000
75%	12.018	18600,000	15827,306	62,000	10,000	62,000
max	99,420	1541000,000	2750000,000	99,000	43,000	90,000

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The Twelve Firms with Highest Market Cap in 2022



There were 12 firms with market capitalization greater than \$250 billion in 2022.

They are shown here ordered alphabetically.

Apple had the highest market cap

Microsoft was next

Then Amazon

CompanyName	MktCap	ExecName	TDC1	Employees	SIC2
Apple	2750000.0	Timothy Cook	99.420097	164000	36
Amazon	1210000.0	Andrewassy	22.498723	1541000	59
Broadcom	283000.0	Hock E. Tan	60.606971	20000	36
Home Depot	299000.0	Edward Decker	14.515600	500000	52
JPMorgan Chase	408000.0	James Dimon	34.818729	293723	60
Coca-Cola	272000.0	James Quincey	22.332484	79000	20
Meta (Facebook)	633000.0	Mark Zuckerberg	27.110418	86482	73
Microsoft	2370000.0	Satya Nadella	54.946309	221000	73
Pepsico	264000.0	Ramon Luis Laguarta	24.136449	315000	20
Procter & Gamble	360000.0	Jon Moeller	17.716015	106000	28
Visa	489000.0	Alfred Kelly, Jr.	28.103602	26500	60

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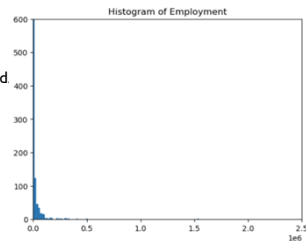
Histogram of Employment



The histogram of employment is highly skewed.

Do histograms of market capitalization and CEO compensation.

- You will find that they are also highly skewed

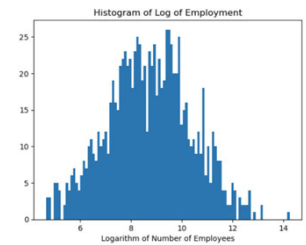


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Histogram of Logarithm of Employment



The histogram of the logarithm of employment is relatively symmetric.



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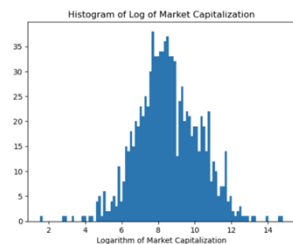
Histogram of Logarithm of Market Capitalization



The histogram of the logarithm of market capitalization is relatively symmetric.

Do a histogram of the logarithm of CEO compensation and you will find it is also relatively symmetric.

These findings regarding distributions of the variables suggest doing a regression using logarithms of variables.



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CEO Pay Regression



Estimate regression using logarithms.

```
PayReg2=lm(log(TDC1)~log(Employees)+log(MktCap), usFirms)
PayReg2.summary()
```

Are the coefficients of the variables significant at the 5% level?

Do the coefficients have the anticipated signs?

Interpret the coefficients.

A one percent increase in the number of employees increases CEO pay by one tenth of a percent holding market capitalization constant.

A one percent increase in the market capitalization increases CEO pay by one fourth of a percent holding number of employees constant.

OLS Regression Results					
Dep. Variable:	log(TDC1)	R-squared:	0.451		
Model:	OLS	Adj. R-squared:	0.450		
No. Observations:	989	F-statistic:	372.1		
Covariance Type:	HC1	Prob (F-statistic):	3.98e-121		
	coef	std err	t	P> t	[0.025 0.975]
Intercept	-1.0981	0.125	-8.786	0.000	-1.343 -0.853
log(Employees)	0.1027	0.016	6.281	0.000	0.071 0.135
log(MktCap)	0.2528	0.015	17.353	0.000	0.224 0.281

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View Leverage Plots

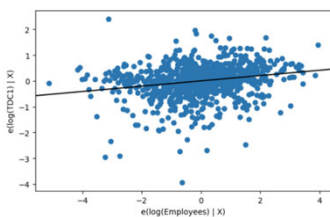


Create leverage plots using:

```
# Display leverage plots
fig = plt.figure(figsize=(12,12))
sm.graphics.plot_partregress_grid(PayReg3, fig=fig)
plt.show()
```

Viewing the leverage plots, the plot relating the log of CEO pay and the log of Employees seemed to me to exhibit some nonlinearity. In particular, near the middle range of the plot, there are more large outliers below the line than above.

Hence, I decided to try including the square of the log of Employees as an additional variable.



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Model Including Square of log of Employees



Viewing the result, we see that the log of employees has p-value =.007 and the square of the log has p-value .061. Hence, the squared term is relatively significant.

The log of market cap continues to be significant.

The remaining two variables continue to be insignificant.

Next slide displays the relationship with respect to the logarithm of employees.

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-2.0230	0.496	-4.082	0.000	-2.995	-1.050
log(Employees)	0.2950	0.109	2.708	0.007	0.081	0.509
l(log(Employees)**2)	-0.0112	0.006	-1.879	0.061	-0.023	0.000
log(MktCap)	0.2572	0.015	17.325	0.000	0.228	0.286
log(YRS_CEO)	0.0303	0.028	1.072	0.284	-0.025	0.086
AGE	0.0006	0.004	0.165	0.869	-0.007	0.008

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Variation of Pay with Number of Employees



This plot show the relationship of CEO pay to number of employees holding other variables constant.

We see that the log of pay increases at a decreasing rate and becomes flat at about 13. Exponentiating 13, this corresponds to 442 thousand employees. The slight dip at the right is likely an artifact of the quadratic approximation and not a reflection of decline in pay.



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Standard Industrial Classification Codes



Each US firm is classified by the US government with a four-digit SIC code.

- The first two digits of the code are the industry in which the firm operates.
- The next two digits denote the particular focus of the firm. For our regression analysis below, we will use the two-digit codes.

As an example, consider Apple:

- The primary four-digit code for Apple is 3663. The first two digits of this code, 36, tell us that the industry in which Apple is classified is "Electronic and Other Electronic Equipment".
- The four-digit code 3663 is: "Radio and Television Broadcasting and Communications Equipment". This classification is obviously due to Apple's focus on Communications Equipment.
- A second SIC code for Apple is 3571, which is "Electronic Computers."

The complete list of codes is at this link:

<https://www.bls.gov/oes/special-requests/oessic87.pdf>

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CEO Pay Variation by Industry

Include indicator variables for Industry
Let's make industry 36 the reference by switching its number to 1
USFirms["SIC2"] = USFirms["SIC2"].replace(36, 1)
PayReg=statsmodels.model("ols", "log(TOBI)-log(Employees)+2*log(MktCap)+log(YRS_CEO)+AGE+C(SIC2)", USFirms)
Now restore SIC value for industry 36
USFirms["SIC2"] = USFirms["SIC2"].replace(1, 36)
PayReg.summary()

The industry with the largest coefficient, .5630, is 15.

Check the codes:
<https://www.bls.gov/oes/special-requests/oessic87.pdf>

C 15 150 1500 Building Construction General Contractors and Operative Builders

The industry with the lowest coefficient, -.5007, is 42.

E 42 420 4200 Motor Freight Transportation and Warehousing

On the next slide, we calculate the implications of the above two values.

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-2.2161	0.546	-4.062	0.000	-3.287	-1.146
log(Employees)	0.3292	0.118	2.799	0.005	0.098	0.560
2*log(MktCap)	-0.0119	0.006	-1.862	0.063	-0.025	0.001
log(YRS_CEO)	0.2128	0.019	13.316	0.000	0.216	0.290
AGE	0.0177	0.028	0.632	0.528	-0.037	0.073
C(SIC2)(T10)	-0.0806	0.112	-0.720	0.472	-0.300	0.139
C(SIC2)(T12)	0.3833	0.234	1.639	0.102	-0.076	0.842
C(SIC2)(T13)	0.2718	0.125	2.179	0.030	0.027	0.517
C(SIC2)(T14)	0.3292	0.164	2.010	0.045	0.008	0.651
C(SIC2)(T15)	0.5630	0.154	3.646	0.000	0.260	0.866
C(SIC2)(T16)	-0.2163	0.253	-0.855	0.393	-0.713	0.280
C(SIC2)(T17)	-0.3039	0.115	-2.640	0.008	-0.530	-0.078
C(SIC2)(T20)	-0.0457	0.123	-0.372	0.710	-0.287	0.195

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Calculating Pay Differences by Industry

On the previous slide, we found Industry 15 had a coefficient of .5630.
The dependent variable is in log form. Hence, the coefficients of the indicator variables for industry are semi-log coefficients.
Recall that we calculate the predicted effect on the dependent variable using:

Calculate Differences Across Industry
Highest
100*(numpy.exp(.5630)-1)

We obtain 75.5932. Thus, other variables held constant, CEO pay in industry 15 is predicted to be 75.6% higher than in industry 36.

Calculate Differences Across Industry
Lowest
100*(numpy.exp(-.5007)-1)

We obtain -38.3894. Thus, other variables held constant, CEO pay in industry 2 is predicted to be 38.4% lower than in industry 36.

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Assessing Functional Form

Prepare and view a plot of residuals versus fitted values:

```
# Extract residuals and fitted values and do plot
resid=PayReg.resid
yhat=PayReg.fittedvalues
plt.scatter(yhat,resid)
plt.xlabel("yhat",label="Fitted Values")
plt.ylabel("resid",label="Residuals")
plt.title("Plot of residuals versus fitted values for CEO Pay Regression")
plt.show
```

There are large outliers but no visual evidence of a nonlinear pattern. There is clear evidence of heteroskedasticity.

The Ramsey test yields relatively high p-values suggesting that this is a suitable functional form for these data.

	power	pvalue
2	0.198944	
3	0.406056	

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Summary

My goals with this application were two-fold.

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